

Adversarial Defense via PGD- Based Adversarial Training

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Motivation:

A small perturbation negatively impacts the model accuracy, is there a way to make the model more robust?

Main Idea:

Adversarial training using PGD generated training data.

Result:

PGD Adversarial Training reduces ASR from 100% → 6.69% and achieves 91.72% robust accuracy with <1% clean accuracy loss.

Literature Review

Intriguing
Properties of
Neural
Networks
Szegedy et al. (2014)

First to discover
adversarial examples.
Invisible perturbations
cause confident
misclassification.
Motivated the entire
field.

Explaining and
Harnessing
Adversarial
Examples
Goodfellow et al. (2015)

Proposed FGSM —
single-step attack.
Vulnerability stems
from the linear nature
of neural networks.
First basic adversarial
training idea.

Towards Deep
Learning Models
Resistant to
Adversarial
Attacks
Madry et al. (2018)

Robustness as min-
max optimization.
PGD = strongest first-
order attack.
PGD adversarial
training is the
gold standard defense.

Approach

- Dataset: MNIST
- Model: small CNN
- Eval attacks: FGSM + PGD (40 steps, $\alpha = \epsilon/10$)
- Optimizer: Adam, lr = 1e-3
- Defense implemented following **Madry et al. 2018** min-max PGD adversarial training framework. Attack baselines include FGSM (**Goodfellow 2015**) and PGD.
- Code: <https://github.com/yang330811/ai-mrp-spring-2026-pgd-at>

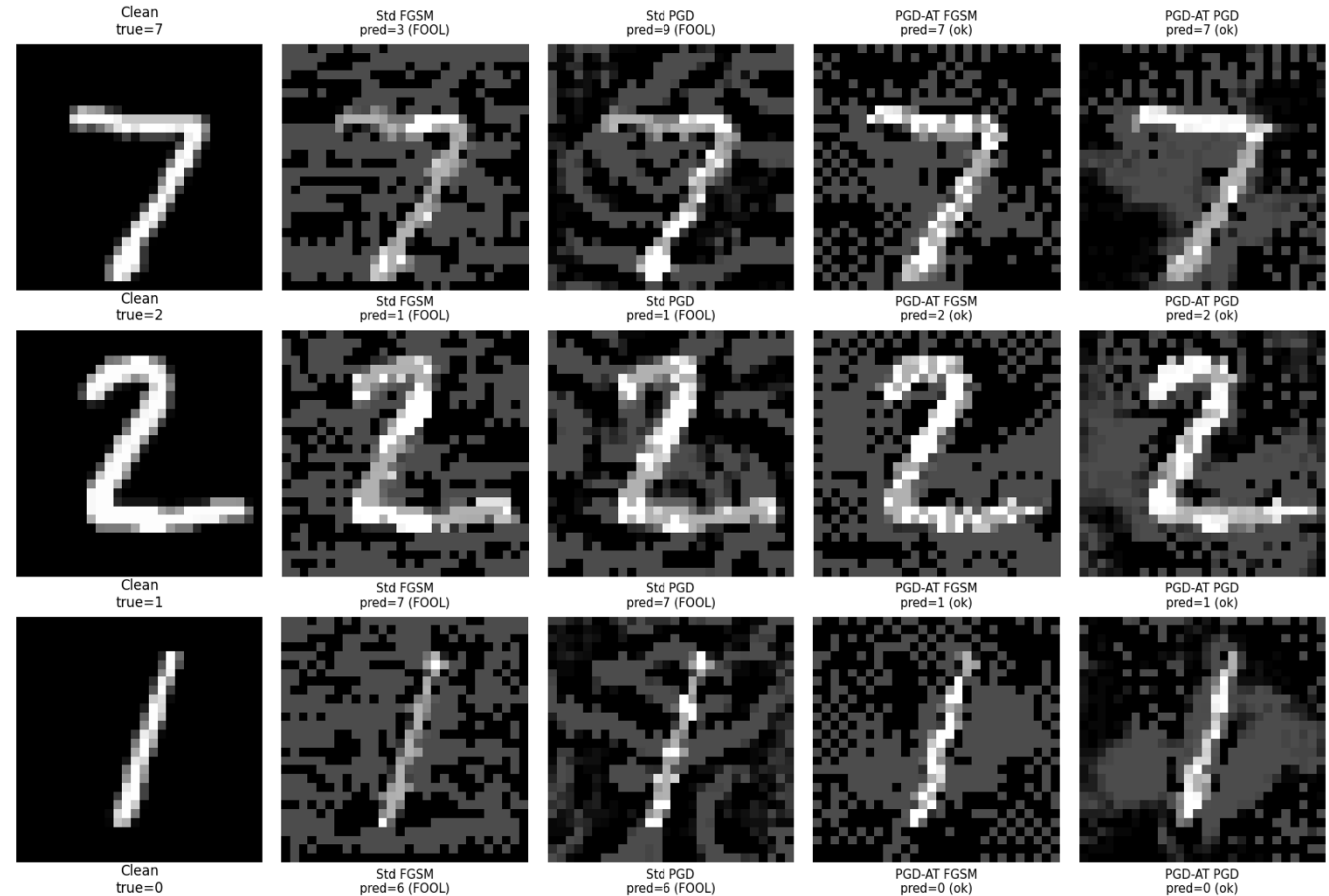
Iteration	ϵ	PGD train steps	Step size α	Epochs	Notes
iter1	0.3	7	0.01	10	default baseline
iter2	0.3	20	0.03 ($\epsilon/10$)	20	stronger PGD training
iter3	0.2	20	0.02 ($\epsilon/10$)	20	smaller perturbation budget

Results

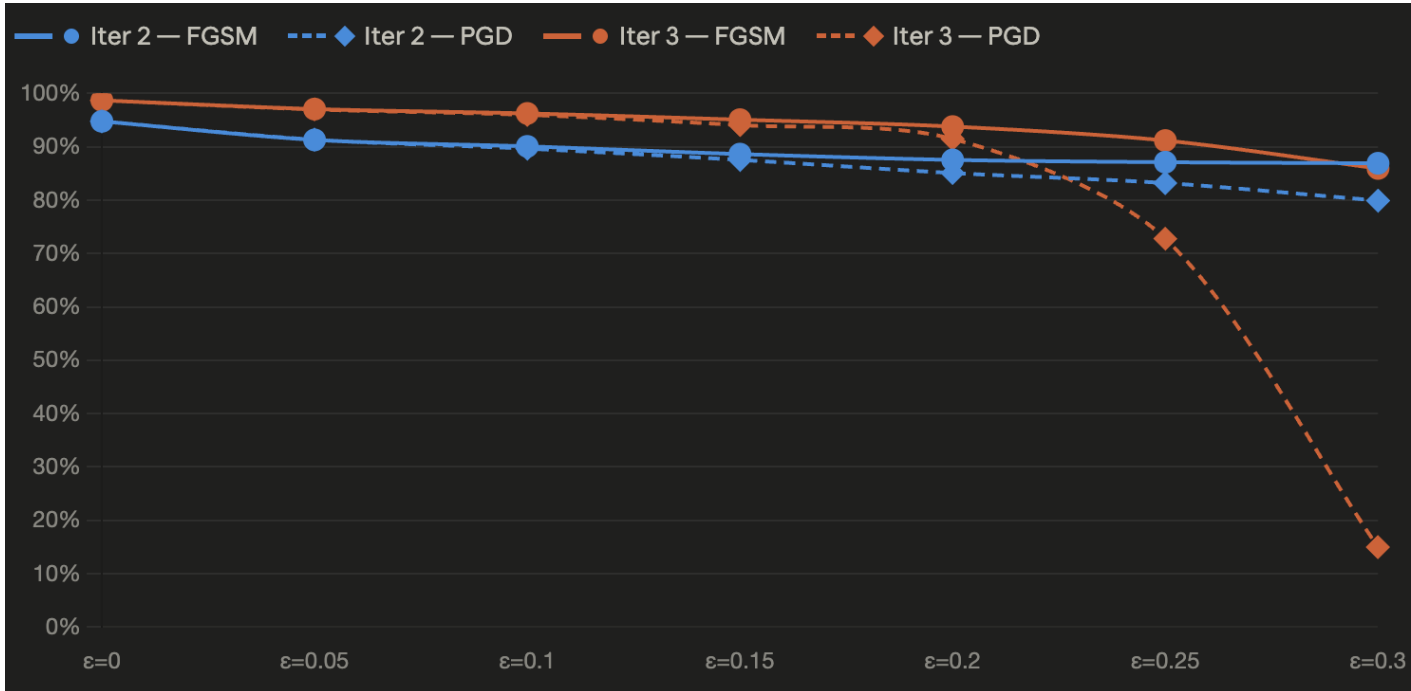


- PGD Adversarial Training reduces Attack Success Rate (ASR) largely from 100% to 6.69%
- More training steps + smaller $\epsilon \rightarrow$ more robust model
- 91.72% robust accuracy with <1% clean accuracy loss

Model Comparison: Standard vs PGD-AT (epsilon=0.3)



Results



Iter 2 & 3 : Robustness vs. epsilon comparison

- Models are only robust within their training ϵ
- Iter 3 ($\epsilon=0.2$) outperforms iter 2 ($\epsilon=0.3$) within training range, but iter 2 holds up better at high ϵ
- Robustness-accuracy trade-off: stronger training distribution = better defense, but only up to what the model has seen

Demo Video

<https://www.youtube.com/watch?v=wVOiLfzmXhE>

Reference

- [1] Szegedy, C. et al. (2014). Intriguing properties of neural networks. ICLR 2014. arxiv.org/abs/1312.6199
- [2] Goodfellow, I., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. ICLR 2015. arxiv.org/abs/1412.6572
- [3] Madry, A. et al. (2018). Towards deep learning models resistant to adversarial attacks. ICLR 2018. arxiv.org/abs/1706.06083